



AI-Based App to screen Cervical cancer

Scientists are making app to use artificial intelligence to identify precancerous or cancerous cells with just a photograph. <https://digit.in/apr19-11>



Samsung patents an "Air quality" app

Samsung filed trademark "SmartThings Air" app for home automation, including lighting, heating, AC & safety equipment. <https://digit.in/apr19-12>

IS IT A CURE?

Neither of the people was being treated for HIV via the stem cell transplant, which was actually intended to combat their cancer. The delta32 mutation is found in about 16% Northern Europeans, and it was present in both of their donors. The nature of bone marrow transplants makes this unlikely as a cure, as the procedure carries its own set of risks. Additionally, multiple attempts to repeat the success of the first instance have failed until the London Patient turned that around. The problem with calling the recent instance as a cure is also rooted in our understanding of the difference between cure and remission.

A cure is when there are no traces of the virus or the disease in the body of the patient whatsoever, and that it will never come back again. A remission is when the presence and the effects of the virus are untraceable in the host, but it still might come back in the future. As mentioned earlier, researchers encountered multiple failures trying to repeat Timothy Brown's case. Each time, the virus would come back once the patients stopped taking their prescribed medicine.

In case of the London Patient, they've been unable to find any traces of HIV even with extraordinarily sensitive tests 18 months after they stopped taking their HIV medicine. While some might see this as cause for celebration, other scientists are not so sure. "We don't have any international agreement on what time without viral rebound is necessary to speak about cure," Dr. Annemarie Wensing, a virologist at the University Medical Center Utrecht in the Netherlands, tells the New York Times.

On the other hand, the bone marrow being solely responsible for the cure is also an aspect that some have raised questions about. For starters, there have been cases where patients have gone into remission without a bone marrow transplant. Additionally, in both the Berlin and London patient cases, something known as graft-versus-host disease was experienced.

In this situation, immune cells from the donor attack immune cells in the host, which would lead to a loss of HIV infected immune cells in the host.

Lastly, the procedure of bone marrow transplant cannot be classified as a cure for HIV due to its inherent risk. Patients undergoing it are at the risk of life threatening complications, mainly involving infections that arise from the immune suppression



Timothy Brown, the Berlin Patient

required for the process. During their treatment, the Berlin patient nearly died and the London patient had to suffer serious infections.

WHY ARE WE LOOKING?

The big question that arises here is that if a person is living a healthy lifestyle, with no complications or effects of HIV affecting them or people around them, as long as they have a particular medicine everyday, is it really worth undergoing a life threatening procedure to get rid of the condition? The answer to that lies partly in the stigma associated with the condition itself. People are often discriminated against and kept from multiple opportunities and avenues in their lives due to being an HIV patient. A relief from this drives people towards any potential cure, and the prospect of keeping it under control via regular medication is always seen as something beneath an actual cure.

Additionally, the current situation is far from ideal. "There are 36 million-odd people with HIV worldwide", Mr Prof Ravindra Gupta from University

College London, the lead author of the paper on the successful treatment of the London patient, tells The Guardian, "The aim is to get everybody on treatment for the rest of their lives and that's a huge undertaking both for drug delivery but also making sure people can stay on medication for decades. There is a cost issue for developing countries."

WHAT'S NEXT

The achievement of science with the Berlin patient twelve years ago and the London patient now shouldn't be seen as path breaking cures by themselves. Instead, they've opened up the doors and reaffirmed the potential for more potent, less risky and possibly affordable methods like gene editing, particularly focused on the CCR5 receptor. This has been done with haemophilia and sickle cell disease.

A study (<https://digit.in/hivstudy>) published in the New England Journal of Medicine details a method to edit extracted CCR5 in the lab and infuse it back into a patient's body. In another trial funded by Sangamo Therapeutics, the amount of virus in the body seemed to decline by a thousand times. Many teams are also looking at the possibility of delivering targeted enzymes to the body that carry out the gene editing within the cells themselves, perhaps via nanoparticles. A group in China is also trying to use modified stem cells (<https://digit.in/cn-hiv>). Around the world, there are many parallel efforts to find a cure.

Even if these theoretical treatments do work out in the next 5-10 years, it would only work against one type of HIV that depends on CCR5 receptors. Another type, called X4, relies on a different protein to enter cells; none of these theoretical treatments would guard against infection with that form of the virus. In either case, regular medication at the moment is the best option until some genetic scientists around the world figure out a way to achieve with gene editing what was achieved for the Berlin and London patients via bone marrow transplants. 